PYTHON FUNDAMENTALS

Assignment — Intermediate Level

20 Problems | Modules: Variables · Operators · Strings · I/O · Conditionals

Student Name: Date:

Instructions

- Solve all 20 problems in Python. Submit as .py files or a single Jupyter Notebook.
- Each solution must run without errors and produce the correct output.
- Add a short comment (1–2 lines) above each problem explaining your approach.
- For problems with user input, test with at least 2 different inputs.
- Refer to the examples notebook (Python_Fundamentals_Examples.ipynb) to review any concept.

Module: Variables & Data Types

Problem 1: Declare five variables: your name (str), age (int), height in meters (float), whether you are a student (bool), and a variable with no value (NoneType). Print all five values in a single print() statement.

Hint: Use name, age, height, is_student, nothing = None. One print() with all five.

Problem 2: Given x = '42', y = 3.9, z = True - convert x to int, y to int (notice what happens to the decimal), and z to str. Print each converted value and verify its type using type().

Hint: int('42'), int(3.9) truncates the decimal. Use type() to confirm.

Problem 3: Assign the values 10, 20, 30 to three variables a, b, c in a single line. Then swap a and c without using a temporary variable and print both before and after values.

Hint: Python allows: a, c = c, a in one line.

Problem 4: Show dynamic typing in action: create a variable x, assign an integer to it and print its type, then reassign a string to the same variable and print its type again, then reassign a float and print once more.

Hint: x = 10 → x = 'hello' → x = 3.14. Use type(x) after each.

Problem 5: Try type-converting the following and print the result: bool(0), bool(''), bool('hello'), bool(99), int(True), int(False), float('3.14'). Write one conversion per line with a comment explaining the output.

Hint: 0 and empty string are falsy. Any non-zero / non-empty is truthy.

Module: Basic Operators

Problem 6: Store any two integers in variables a and b. Print the result of all seven arithmetic operations (+, -, *, /, //, %, **) with a descriptive label for each line. Example: 'Floor division: 3'.

Hint: Use f-strings or string concatenation for the labels.

Problem 7: Start with x = 100. Apply these augmented assignment operations in order and print x after each step: subtract 25, multiply by 2, floor-divide by 3, take modulus with 7, raise to power 2.

*Hint: x -= 25, then x *= 2, etc. Print after every operation.*

Problem 8: Given a = 15, b = 15, c = [1,2], d = [1,2] - print the result of: a == b, a is b, c == d, c is d. Then explain in a comment why 'is' and '==' give different results for the lists.

Hint: == checks value equality, 'is' checks if both point to the exact same object in memory.

Problem 9: Write expressions using logical operators to check: (1) a number is between 10 and 50 (inclusive), (2) a string is either 'yes' or 'no', (3) a number is NOT divisible by 2. Test each with sample values and print True/False.

Hint: Use 'and', 'or', 'not', and the % operator.

Problem 10: Check membership using 'in' and 'not in': (1) is 'py' inside 'python'? (2) is 'z' not in 'python'? (3) is 7 in the list [1, 3, 5, 7, 9]? (4) is 4 not in the same list? Print each result.

Hint: Both strings and lists support 'in' and 'not in'.

Module: String Manipulation

Problem 11: Take the string s = ' python programming '. In separate print statements, show: the stripped version, stripped + upper(), stripped + title(), stripped + replace('python', 'data'), and the length of the stripped string.

Hint: Call .strip() first then chain other methods, or store stripped version in a variable.

Problem 12: Given text = 'Data,Science,Machine,Learning' - split it into a list, print the list, print the second element, print the total number of elements, then join the list back with ' → ' as separator.

Hint: Use split(','), len(), and ' → '.join(list).

Problem 13: Given sentence = 'Hello World Python' - use slicing to print: the first 5 characters, the last 6 characters, every second character, and the entire string reversed.

Hint: [:5], [-6:], [::2], [::-1]

Problem 14: For the string word = 'Banana' - print the result of: find('a'), count('a'), startswith('Ba'), endswith('na'), isdigit(). Then check if '12345'.isdigit() is different from '123a5'.isdigit().

Hint: find() returns the index of first occurrence. isdigit() returns False if any char is not a digit.

Problem 15: Create three variables: product = 'Laptop', price = 75000.5, discount = 10. Print a formatted line using all three string formatting styles (f-string, .format(), % operator) showing: 'Product: Laptop | Price: 75000.50 | Discount: 10%'.

Hint: Use :.2f for price formatting in all three styles.

Module: Input / Output

Problem 16: Write a program that takes the user's name and birth year as input. Calculate their approximate age (current year - birth year) and print: 'Hello [name]! You are approximately [age] years old.'

Hint: birth_year = int(input(...)). Use 2025 as current year or hardcode it.

Problem 17: Build a simple area calculator. Ask the user to enter length and width (as floats). Calculate area and perimeter of the rectangle. Print both results formatted to 2 decimal places with descriptive labels.

*Hint: area = l * w, perimeter = 2 * (l + w). Use float(input(...)).*

Module: Conditional Statements

Problem 18: Take a score (0 - 100) from the user and print a grade using if-elif-else: 90–100 → 'A+', 80–89 → 'A', 60–79 → 'B', 40–59 → 'C', below 40 → 'F'. Also print 'Pass' or 'Fail' based on score >= 40.

Hint: Use 5-branch if-elif-else for grade, then a separate if-else for Pass/Fail.

Problem 19: Write a login system. Store a correct username and password in two variables. Take username and password input from the user. Using nested if-else: if username matches → check password → print 'Access Granted' or 'Wrong Password'; if username doesn't match → print 'User Not Found'.

Hint: Outer if checks username. Inner if checks password only if username is correct.

Problem 20: Take any number from the user. Using a single line ternary expression, print whether it is 'Positive', 'Negative', or 'Zero'. Then separately, using nested if-elif-else, also check if the number is even or odd (only if it is not zero).

Hint: Ternary: 'Positive' if n > 0 else ('Negative' if n < 0 else 'Zero'). Nested for even/odd.

Good luck!